

Health Risk Modeling in CDC Survey Data

A LASSO-Based Classification Study Using BRFSS 2024

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Executive Summary

This report presents the findings of a health risk modeling study using the CDC Behavioral Risk Factor Surveillance System (BRFSS) 2024 survey data. The project involved building two interpretable binary classification models to identify key predictors of heart disease risk and severe poor physical health among U.S. adults.

The analysis applied LASSO-regularized logistic regression to a dataset of 457,670 survey respondents with 301 raw variables, narrowed to approximately 40 clinically meaningful predictors. General health status emerged as the strongest predictor across both models, with chronic conditions, socioeconomic factors, and healthcare access also playing significant roles.

Key Findings at a Glance

Finding	Detail
Dataset	457,670 BRFSS 2024 records; ~40 selected predictors from 301
Model 1: Heart Disease	19 LASSO-selected predictors; AUC improved via threshold tuning
Model 2: Severe Poor Health	10 LASSO-selected predictors; strong mental health signal
Top Predictor (Both Models)	General health status (GENHLTH)
Heart Disease Sensitivity	Improved from 12.4% (default) to 50.0% (tuned threshold)
Class Imbalance	Only 9% of cases had heart disease — accuracy alone is misleading

Background & Motivation

Heart disease remains one of the leading causes of mortality in the United States. Cardiovascular disease accounts for approximately 1 in 5 deaths annually, with one person dying every 34 seconds from cardiovascular disease. The financial burden is equally staggering — health care services and medications for heart disease amounted to more than \$168 billion between 2021 and 2022.

Clinical prediction models must be both interpretable and actionable. This project focuses on identifying meaningful health risk factors from large-scale population survey data to support early detection, prevention, and equitable healthcare planning.

Data: CDC BRFSS 2024

The Behavioral Risk Factor Surveillance System (BRFSS) is the world's largest continuously conducted health survey. The 2024 edition provides comprehensive data on U.S. adult health behaviors, chronic conditions, and healthcare utilization.

Survey Records	Raw Variables	Selected Predictors
457,670	301	~40

Variable Domains

Domain	Variables Included
Health Outcomes	GENHLTH (general health), PHYSHLTH, MENTHLTH
Heart Disease	CVDINFR4 (heart attack), CVDCRHD4 (coronary heart disease)
Sociodemographic	Sex, age group, education, income, employment, marital status
Lifestyle	Exercise (EXERANY2), smoking (SMOKE100, SMOKDAY2), e-cigarettes (ECIGNOW3)
Clinical	BMI, diabetes, stroke, kidney disease, depression, arthritis, COPD, asthma
Healthcare Access	Insurance (PRIMINS2), personal doctor (PERSDOC3), cost barriers (MEDCOST1), last checkup (CHECKUP1)
Disability / Mobility	DIFFWALK, DIFFALON, DEAF, BLIND, DECIDE

Methodology

The analysis followed a rigorous five-step pipeline designed to maximize interpretability while handling the complexities of large health survey data.

Step 1: Data Cleaning

BRFSS uses special response codes to indicate refusals, don't-know responses, and not-applicable cases. These were systematically recorded as missing values:

- Codes 7, 9, 77, 99 treated as NA (refused / don't know)
- Code 88 in PHYSHLTH and MENTHLTH converted to 0 days ("none" response)

Step 2: Target Variable Creation

- Model 1 — Heart Disease: Binary indicator (1 = heart attack OR coronary heart disease; 0 = neither)
- Model 2 — Severe Poor Health: Binary indicator for reporting 30 days of poor physical health per month

The original poor_health_days variable was heavily right-skewed with large spikes at 0 and 30, violating linear regression assumptions and motivating the shift to a binary outcome.

Step 3: Feature Selection

Administrative and survey-processing variables were removed, retaining 40 clinically meaningful predictors. Non-response indicators (nonresp_PHYSHLTH, nonresp_MENTHLTH, nonresp_GENHLTH) were also created, as missingness patterns can carry predictive signal.

Step 4: LASSO Variable Selection

Cross-validated LASSO logistic regression was applied using lambda.1se (the more conservative, parsimonious choice) rather than lambda.min. This approach balances predictive accuracy with model simplicity, shrinking irrelevant predictors to zero. Data was split 60/20/20 into training, test, and calibration sets.

Step 5: Final Model Evaluation

Final logistic regression models were refit using only LASSO-selected predictors and evaluated on held-out test data using:

- Confusion matrix (accuracy, sensitivity, specificity)
- Area under the ROC curve (AUC)
- Odds ratios with interpretation

- Threshold tuning (for heart disease model) to improve sensitivity

Project Evolution

The original project plan used linear regression on the continuous `poor_health_days` outcome combined with conformal prediction intervals. During exploratory analysis, three critical issues emerged that necessitated a redesign:

- **Distribution problems:** `poor_health_days` had severe bimodal spikes at 0 and 30 days, making linear regression assumptions untenable.
- **Data leakage:** Using `PHYSHLTH` to predict `poor_health_days` essentially amounted to predicting a variable using itself — a fundamental validity flaw.
- **Interpretability:** Conformal prediction intervals added mathematical complexity while limiting practical interpretability and clinical actionability.

The revised approach — two binary LASSO-selected logistic regression models — addressed all three issues while remaining accessible and clinically interpretable.

Model 1: Heart Disease Risk Classification

Overview

Model 1 predicts binary heart disease status (yes/no) using cross-validated LASSO logistic regression with `lambda.1se`. The final model retained 19 of the ~40 candidate predictors, effectively shrinking more than half to zero.

Key Predictors & Interpretation

Predictor	Odds Ratio & Interpretation
GENHLTH (General Health)	OR = 1.54 — Each one-category worsening in self-rated health (e.g., Good → Fair) increases odds of heart disease by ~54%
_AGEG5YR (Age Group)	OR = 1.26 — Older respondents have substantially higher heart disease risk
DIFFWALK (Difficulty Walking)	Strong predictor — signals reduced mobility and cumulative health burden
nonresp_GENHLTH	Non-response itself carries predictive signal — missingness may reflect avoidance or disengagement from healthcare
CHECKUP1 (Regular Checkups)	Earlier detection and preventive care associated with better outcomes
PERSDOC3 (Personal Doctor)	Having a consistent primary care provider linked to reduced risk
INCOME3 (Income Level)	Lower income associated with higher cardiovascular risk
EMPLOY1 (Employment Status)	Employment status captures socioeconomic and activity-level signals

Summary: Heart disease risk is driven by a combination of clinical deterioration markers, functional limitations, socioeconomic disadvantage, and reduced healthcare engagement.

Threshold Tuning for Medical Screening

In medical screening contexts, false negatives (missing a true disease case) are far more costly than false positives (flagging a healthy patient). The default classification threshold of 0.50 produced very high accuracy but unacceptably low sensitivity.

Metric	Default Threshold (0.50)	Tuned Threshold (0.20)
Accuracy	91.1%	86.5%
Sensitivity	12.4%	50.0%
Specificity	98.9%	90.2%

By lowering the threshold to 0.20, the model accepted more false positives in exchange for detecting four times as many true heart disease cases, a clinically justified tradeoff for a screening tool.

Model 2: Severe Poor Health Burden Classification

Overview

Model 2 predicts whether a respondent reports 30 days of poor physical health per month — a marker of chronic, severe health burden. LASSO with lambda.1se selected 10 predictors from the candidate set.

Key Predictors & Interpretation

Predictor	Odds Ratio & Interpretation
GENHLTH (General Health)	Strongest predictor — self-rated health captures overall well-being and chronic disease burden
_AGEG5YR (Age Group)	OR = 1.26 — Older age strongly associated with chronic poor physical health
MENTHLTH (Mental Health Days)	OR = 1.03 per additional day — each additional poor mental health day increases odds of severe poor physical health by ~3%
EXERANY2 (Exercise)	OR = 1.53 — Lack of regular exercise strongly associated with worse physical health outcomes
nonresp_MENTHLTH	Missingness in mental health responses also carries predictive signal
Chronic Conditions	Stroke history, kidney disease, COPD, and mobility limitations were among the selected predictors

A notable finding: mental health emerged as a significant and independent predictor of severe poor physical health. This reinforces the importance of integrated mental and physical healthcare approaches.

Cross-Model Findings & Key Insights

Consistent Predictors Across Both Models

- General health status (GENHLTH) was the strongest predictor in both models, underscoring its value as a simple, powerful screening indicator.
- Age group (_AGEG5YR) showed consistent associations with both heart disease and severe poor physical health.
- Chronic conditions — including stroke, kidney disease, COPD, diabetes, and mobility limitations — were major drivers in both outcomes.
- Non-response patterns (nonresp_GENHLTH, nonresp_MENTHLTH) carried predictive information, suggesting that survey missingness is not random.

Socioeconomic and Access Factors

Income, employment status, access to a personal doctor, and frequency of preventive checkups all contributed meaningfully to heart disease predictions. These findings point to health disparities that extend beyond clinical risk factors alone.

Mental Health as a Physical Health Predictor

The inclusion of MENTHLTH in the severe poor health model highlights that mental and physical health are deeply intertwined. Ignoring mental health in physical risk assessments likely leads to underestimation of health burden.

Limitations & Future Directions

Current Limitations

- Severe class imbalance (9% heart disease cases) reduced sensitivity even after threshold tuning.
- BRFSS survey weights were not incorporated, meaning results are not nationally representative.
- Data is self-reported and subject to recall bias and social desirability effects.
- The bimodal distribution of poor_health_days limited the regression approach and required the binary outcome redesign.

Future Improvements

- Implement SMOTE or cost-sensitive learning algorithms to better handle class imbalance.
- Test ensemble methods such as Random Forest or XGBoost for improved predictive performance.
- Incorporate BRFSS complex survey weights for valid population-level inference.
- Conduct subgroup fairness analysis by sex, race/ethnicity, and age to identify disparate model performance.
- Explore interaction terms between socioeconomic and clinical predictors.

Practical Recommendations

The findings from this analysis translate into actionable public health and clinical guidance:

Recommendation	Supporting Evidence
Maintain regular physical activity	Difficulty walking and lack of exercise were strong predictors in both models — mobility maintenance is critical

Proactive chronic condition management	Diabetes, stroke history, kidney disease, COPD, and arthritis all correlate with worse outcomes
Prioritize preventive healthcare	Regular checkups and having a personal doctor were associated with better outcomes and earlier intervention
Address mental health proactively	Mental health burden was an independent predictor of severe poor physical health — treat both together
Reduce socioeconomic health barriers	Income and healthcare access barriers independently predicted heart disease risk — systemic interventions are needed
Improve screening sensitivity	Threshold-tuned models should be used in screening contexts to minimize missed diagnoses

Technical Appendix: Modeling Pipeline

The analysis was implemented in R using the following primary packages:

- haven — BRFSS XPT file loading
- dplyr — data manipulation and variable creation
- glmnet — cross-validated LASSO logistic regression
- caret — confusion matrix and performance metrics
- ggplot2 — exploratory visualization

Data Split

Split	Proportion	Purpose
Training	60%	Model fitting and LASSO selection
Test	20%	Final performance evaluation
Calibration	20%	Conformal prediction (early stage)

LASSO Configuration

Both models used `cv.glmnet` with `family = "binomial"` and `alpha = 1` (pure LASSO). Lambda selection used `lambda.1se` rather than `lambda.min` to prefer sparser, more interpretable models. Odds ratios were computed as `exp(coef(logit_model))` on the final logistic regression refit.

Data source: CDC BRFSS 2024 | <https://www.cdc.gov/brfss/>